



New Draft Proposal

Challenge: Human Survivability

PROPOSAL TITLE * ()

Please provide the title of the technical solution you are proposing.

Hybrid propulsion system VTOL drone with tilt rotor system for medical and defence applications capable of long-range deliveries



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DynoBay

TRL LEVEL * ()

Please select the Technology Readiness Level ("TRL") of the technical solution you are proposing to the challenge.

TRL 4 Technology validated in lab

SYSTEM TYPE * ()

State whether your technical solution is software, hardware or a combination of both.

Hardware

SYSTEM LEVEL * ()

State whether your technical solution is by itself a platform or system, or whether it is a sub-system or component that would integrate into another system or platform.

System

Short form proposal

PART 1 - ALIGNMENT TO CHALLENGE * ()

- What elements of the challenge are you addressing?
- How will your technical solution address the challenge?
- What is novel and or/unique about your solution?
- What defence and security needs are addressed by your technical solution?

1. Elements of challenge:

- Delayed or denied evacuation due to contested ground and airspace.

PART 2 - TECHNICAL APPROACH * ()

- Describe your technical solution. You may attach graphics or images to help convey your concept.

1. Full-scale concept:

- The full-scale OrgoFly concept uses a tilt-rotor VTOL architecture combining runway-independent access with efficient fixed-wing cruise for medical logistics.
- For take-off and landing, rotors point vertically to



- Overloaded local and rear medical capacity in high-tempo operations.
- Gap between point of injury, stabilisation, and higher care.
- Need to support dispersed units without exposing more personnel.

2. How OrgoFly addresses it:

- Delivers medical and survivability payloads directly to units and field-care points.
- Supports stabilisation and prolonged field care until evacuation is possible.
- Reduces reliance on roads, runways, convoys, helicopters, and large logistics nodes.
- VTOL enables austere launch/recovery; fixed-wing flight provides useful range.

3. Novelty/uniqueness:

- Tilt-rotor architecture uniquely combines VTOL access with efficient forward flight.
- This enables a stronger payload/range fit for contested medical resupply than fixed-rotor VTOL drones.
- Temperature-controlled bay with active cooling supports sensitive medicines, diagnostics, blood, plasma, countermeasures, trauma kits, and warming equipment.
- Modular payload bay enables rapid reconfiguration between different medical and survivability payload kits on a single aircraft.

4. Defence/security and dual-use needs:

- Forward medical resupply and resilient logistics for dispersed units.
- Reduced personnel exposure during dangerous resupply missions.
- Support to prolonged field care when evacuation is delayed.
- Medical delivery directly to the point of injury and Role 1 and Role 2 hospitals without dedicated infrastructure.

PART 3 - TECHNOLOGY DEVELOPMENT ROADMAP WITH RISKS AND MITIGATIONS * ()

- How will the challenge funding help you to mature your proposed technical solution into a demonstrable and/or deployable technology?
- Key milestones:
 - Six months: What do you plan to achieve in six months?
 - One year: If selected for Mission Track and further challenge funding were offered, what would you intend to achieve in one year?
- Risks: What are the most significant risks associated with achieving these milestones?
- Mitigation: What measures are you proposing to mitigate these risks?

1. Funding use:

- DIANA funding supports the 1-year phase after the

generate lift from confined or improvised areas.

- After climb-out, rotors rotate forward, and the aircraft transitions into wing-borne cruise.

2. Airframe and propulsion:

- Unlike VTOL UAVs with separate lift rotors and a pusher motor, OrgoFly uses the same rotors for hover and forward flight.
- This adds complexity, but reduces redundant propulsion hardware and improves aerodynamic efficiency, range, and mission flexibility.
- The full-scale concept includes a folding wing for transport, storage, and deployment from constrained environments.
- Hybrid-electric propulsion provides endurance, energy efficiency, and operational resilience.
- A redundant battery system supports emergency propulsion and auxiliary equipment.

3. Medical payload system:

- The full-scale concept includes a temperature-controlled medical payload compartment for safe medical transport.
- Payload conditions are continuously monitored to support traceability and usability.

4. Operational capability:

- Together, these elements enable runway-independent delivery from field hospitals, remote bases, ships, islands, disaster zones, and improvised landing areas.
- Target full-scale capability is 15 kg payload and up to 400 km range, enabling meaningful medical logistics beyond small package delivery

PART 4 - BUSINESS CASE * ()

- What are the applications of your solution in the defence and security sector? Describe any routes to adoption that you have identified.
- What is the market for your solution?
- What is your current level of investment from public and/or private sources?
- Do you have existing relationships with industry primes? Do you intend to partner with a systems integrator or provide an integrated solution?

1. Defence/security applications:

- Forward medical resupply to dispersed units in contested ground/airspace.
- Delivery of trauma kits, medicines, diagnostics,

bootstrapped 1:4 concept.

- Used for full-scale prototype and certification: folding wing, hybrid propulsion, modular payload bay, temperature-controlled bay, avionics, testing, and BVLOS prep.

2. Six months (bootstrapped):

- Demonstrate full flight capabilities in all operational phases and environments.
- Gather data on stability, range, payload effects, reliability, and scaling.
- Demonstrate successful downsized payload delivery
- Initiate certification process: safety case, telemetry, failsafes, logs.

3. One year: (with funding)

- Build full-scale prototype with hybrid propulsion, folding wing, modular, temperature-controlled cargo bay.
- Demonstrate full defence profile
- Progress BVLOS certification evidence.

4. Risks:

- Civilian certification delays - BVLOS, Specified category, medical
- EW/GNSS degradation, contested airspace, kinetic threats including interceptor drones.
- Evolving EASA/FAA civilian UAV airspace use regulations
- Rapid battlefield evolution may shift mission needs.

5. Mitigations:

- Conduct permitted civilian operations VLOS and focus on defence mission profile.
- Build certification evidence early through tests, logs, failsafes, and regulator/test-site engagement.
- Integrate state-of-the-art navigation and communication components that are modular and upgradeable
- Use modular, upgradable navigation, communications, propulsion, and payload architecture.
- Update mission profiles through user/defence feedback.

countermeasures, blood, plasma, and warming equipment.

- Support to prolonged field care when evacuation is delayed.
- Resilient logistics where convoys, helicopters, runways, or large hubs are unavailable.

2. Market:

- Civilian beachhead: hospital-to-hospital organ and sensitive medical delivery without dedicated infrastructure.
- Organ logistics is a niche but \$600M market, with strong fit covering 80% of missions with our 15 kg / 400 km organ-box.
- Expansion path into transplant-adjacent urgent freight: samples, blood, plasma, diagnostics, and medical consumables.

3. Routes to adoption:

- Use civilian applications to monetize early operations and support validation and certification efforts alongside defence-related development.
- Defence route through GATE Institute – A Partner for Innovation in Security and Defence.
- DIANA would support user validation, testing, and access to defence/security stakeholders.

4. Investment:

- Current development is bootstrapped toward the 1:4 concept.
- DIANA funding would accelerate full-scale prototype development and the certification process.

5. Partnerships:

- GATE Institute is the identified defence/security innovation partner.
- We intend to provide an integrated solution and partner with Tier 1 and Tier 2 defence contractors to tailor to the defence segment

IMAGES

You may upload a maximum of two (2) images containing imagery or graphics. These images or graphics may contain words to signify elements within the images or graphs, however, they must not contain narrative text with explanations or commentary. Any images that are uploaded that attempt to supplement their proposal with additional narrative text may render the proposal non-compliant. Images of block text will be discarded.

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Long form proposal

PART 0 - ONE-SENTENCE DESCRIPTION * ()

Provide an overview of your technical solution in one sentence based on the following format: "the goal is to [provide this capability] using [this technology], to achieve [this impact]."

The goal is to provide point-of-need medical resupply using a hybrid tilt-rotor VTOL aircraft to sustain care across contested/degraded routes.

PART 1: ABSTRACT * ()

Provide a short summary of your solution in one paragraph. Think of this as a written "elevator pitch".

OrgoFly is a hybrid tilt-rotor VTOL aircraft for critical medical logistics in contested and degraded environments. It combines runway-independent launch and recovery with efficient fixed-wing cruise to deliver medical and survivability payloads to dispersed units, Role 1 and Role 2 care points, hospitals, ships, islands, or disaster zones. Its modular, temperature-controlled bay supports sensitive medical payloads and traceability. With a full-scale capability of a 15 kg payload and up to 400 km range, OrgoFly sustains care when roads, helicopters, runways, or rear medical capacity are constrained, ensuring rapid, reliable, and flexible delivery under pressure.

PART 2 - ALIGNMENT TO CHALLENGE * ()

Describe which elements of the challenge your technical solution intends to address. Explain what is novel or unique about your technical solution.

OrgoFly improves access to critical medical supplies when conventional logistics are limited. It is designed for contested or degraded environments where ground routes are unsafe, airspace is restricted, evacuation is delayed, and medical capacity is strained.

The system addresses the gap between point of injury, initial stabilisation, Role 1/Role 2 care, and higher-level medical support. When casualties cannot be moved immediately, medics and field hospitals still need urgent supplies. Our drone delivers critical items such as blood, plasma, medicines, diagnostics, countermeasures, trauma kits, and warming equipment directly to where they are needed.

We do not intend on fully replacing casualty evacuation, but to provide medical resupply that supports stabilisation and prolonged field care until evacuation or advanced care is possible. We supports point-of-injury teams and move urgent supplies between field hospitals, remote bases, ships, islands, and other constrained medical sites. This is especially valuable when demand exceeds medical capacity or when moving supplies is safer or faster than moving patients.

Our competitive edge is in combining VTOL access with the range efficiency of fixed-wing flight in a platform sized for substantial medical payloads. Its tilt-rotor design allows vertical takeoff and landing from remote locations, then transitions to efficient forward flight for long-range resupply. This provides a better payload and range balance for dispersed medical logistics than fixed-rotor VTOL systems.

The full-scale concept features a medical-grade, actively temperature-controlled payload bay for sensitive supplies such as blood, plasma, medicines, diagnostics, and other critical items. This ensures the platform is suitable for medical deliveries where temperature control, traceability, and payload integrity are essential.

With tilt-rotor VTOL access, fixed-wing range, 15 kg-class payload capacity, and active cooling for medical supplies, OrgoFly directly supports sustained care in contested, degraded, or infrastructure-limited environments.

PART 3 - TECHNICAL SOLUTION * ()

Describe your technical solution to the challenge in detail. Describe the current state of development of your solution and outline key performance metrics. Highlight any completed or planned test campaigns.

PART 4 - TECHNOLOGY USE CASE * ()

Provide example use cases of how your technical solution could be applied in defence and security settings, highlighting any direct military use cases. Describe the range of entities (military, civil protection, national security agency etc.) that could potentially adopt your technical solution. You are not expected to have a deep knowledge of defence and security operational uses but should describe some potential applications. Some examples are given in the challenge statement.

PART 5 - BUSINESS CASE * ()

- Describe the potential for the adoption of your technical solution by potential end users, and how this relates to addressable market segments and how you will exploit these. Highlight any routes to adoption that you have already identified, such as through government contracts, defence primes, etc.
- How do you intend to scale your solution? Do you have any intellectual property rights or dependencies that impact the commercial value of the proposed solution? What is your anticipated revenue (including funding opportunities) and current financial runway? Also include any open standards that have been implemented in your solution.

1. Adoption potential:

Initial customers will be medical institutions with urgent local delivery needs, such as maternity and surgical centers that currently dispatch staff or ambulances to collect blood from transfusion centers, resulting in critical delays.

Civilian adoption will begin with operated missions for blood, plasma, samples, and urgent payloads under permitted VLOS operations, followed by BVLOS hospital-to-hospital corridors.

Defence adoption will focus on aircraft sales or operator partnerships for medical resupply to dispersed units, Role 1 and Role 2 field hospitals.

2. Market:

The initial civilian market is urgent medical logistics. Organ transport represents a high-value segment, with a global physical transport TAM of approximately \$600 million annually.

OrgoFly's 15 kg payload capacity, 400 km range, active cooling, and VTOL capability are well-suited for time-critical deliveries where speed, traceability, and the ability to avoid helipads or runways are essential.

Pricing will remain below that of helicopter air ambulances or charter aircraft alternatives, aiming for \$3k inside cities and \$5k-20k for missions 30-400km.

3. Routes to adoption:

The civilian route involves paid pilot programs with hospitals and transfusion centers, followed by certified medical corridors and operators outside the EU.

The defence route includes informal engagement with the GATE Institute through its InnovationAmp support pathway and participation in the NATO DIANA test centre.

We plan to partner with Tier 1 and Tier 2 defence contractors and integrators.

4. Scaling:

Civilian scaling will begin with in-house manufacturing in Bulgaria to maintain cost and quality control.

Expansion beyond the EU will rely on certified operators and local partners.

Defence scaling will begin in-house, with contract manufacturing within permitted borders if speed is a priority.

• Defence units are expected to sell for \$200k-\$500k depending on configuration.

5. IP and dependencies:

Commercial value is derived from proprietary design files, tilt-rotor integration, folding-wing packaging, modular payload architecture, temperature-controlled bays, and specialized know-how.

Key inputs are sourced from multiple EU and US suppliers, reducing dependency on any single supplier.

6. Revenue, funding and runway:

Development is fully bootstrapped, with a runway of nine months.

Near-term revenue is expected from paid civilian pilot programs and permitted VLOS missions.

Additional funding may be secured from DIANA, GATE/InnovationAmp, defence innovation programs, and medical logistics trials.

Medium-term revenue will be generated from recurring civilian missions and defence aircraft sales or integrator partnerships.

7. Open standards:

Open standards will not be claimed until final avionics, telemetry, certification, and payload interfaces are established.

PART 6 - TECHNOLOGY DEVELOPMENT ROADMAP WITH RISKS AND MITIGATIONS * ()

- Provide an estimation of your solution's current technology readiness level ("TRL") for defence applications and, if different, for civilian applications. Describe how the DIANA programme can enable the TRL to advance and how the technical solution can progress towards a deployable capability.
- What is the overall roadmap to achieve an operational solution? What key milestones are achievable in six months? What key milestones are achievable after one year or longer? Identify key risks (technical and otherwise) associated with achieving these milestones and how you propose to mitigate these risks.

PART 7 - TEAM AND RESOURCES * ()

- Describe the background and experience of the personnel in your team and how they are positioned to execute the proposed roadmap and benefit from the DIANA programme. Set out how your team would continue to advance the technical solution and balance core business operations alongside the required programme engagements.
- Describe how you intend to use the €100,000 contractual challenge funding, noting that all funds must be expended in support of the DIANA programme objectives and directly related to the technical solution being proposed.

1. Team:

- Aleksandar Nikolov is an aerospace engineer with aircraft maintenance/MRO, avionics, project planning, composites, and TUM aerospace experience.
- Nikola Nikolov is an aerospace engineer focused on CAD, UAV systems, testing support, sensors, and aircraft maintenance.
- Martin Daskalov is an aerospace engineer in structural dynamics and propulsion, with DLR, University of Stuttgart, EUROAVIA, composite/prototyping, and medical simulation publication experience.
- Martin Valchanov leads business, strategy, operations, partnerships, and programme management, with founder experience at Aver Mobility, PM and consulting work at Medina Med, and PRINCE2 Practitioner certification.

2. Execution fit:

- The team combines aerospace design, propulsion, structures, avionics, CAD, sensors, aircraft maintenance, testing, business development, and project management.
- Members have long-standing school/university links and shared project history, giving strong cohesion and fast coordination.
- Roles split between technical build/testing, certification evidence, partnerships, and commercialization.

3. Use of funds:

- Full-scale prototype development after the bootstrapped 1:4 concept.
- Folding wing, hybrid propulsion, modular payload bay, temperature-controlled bay, avionics, telemetry, failsafes, and test equipment.
- Flight testing, certification evidence, user validation, and mandatory DIANA travel/engagements.
- All funds directly support DIANA objectives and the proposed medical logistics capability.

IMAGES

You may upload a maximum of three (3) images containing imagery or graphics. These images or graphics may contain words to signify elements within the images or graphs, however, they must not contain narrative text with explanations or commentary. Any images that are uploaded that attempt to supplement their proposal with additional narrative text may render the proposal non-compliant. Images of block text will be discarded.

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EMERGING AND DISRUPTIVE TECHNOLOGIES

Please select the NATO Emerging and Disruptive Technologies ("EDT") that apply to your technical solution. Further information can be found: https://www.nato.int/cps/en/natohq/topics_184303.htm

- ☐ Artificial intelligence (AI)
- ☐ Quantum technologies
- ☐ Biotechnology and human enhancement technologies
- ☐ Space
- ☐ Hypersonic systems
- ☐ Novel materials and manufacturing
- ☐ Energy and propulsion
- ☐ Next-generation communication networks
- ☒ Autonomous Systems

TECHNICAL TAGS * ()

Please select any tags that relate to your technical solution. You may select as many tags as you wish.

Select...

AI, Autonomy & Decision Systems > Situational Awareness ✕

Unmanned Systems (Air, Land, Sea) > Unmanned Systems ✕

Unmanned Systems (Air, Land, Sea) > Drone UAV ✕

Unmanned Systems (Air, Land, Sea) > UAS Uxs ✕

Unmanned Systems (Air, Land, Sea) > Autonomous Navigation ✕

Unmanned Systems (Air, Land, Sea) > GPS Denied Navigation ✕

Unmanned Systems (Air, Land, Sea) > Path Planning ✕

Unmanned Systems (Air, Land, Sea) > Obstacle Detection ✕

Unmanned Systems (Air, Land, Sea) > SLAM Mapping ✕

Unmanned Systems (Air, Land, Sea) > Autonomous Control ✕

Unmanned Systems (Air, Land, Sea) > Mission Autonomy ✕

Unmanned Systems (Air, Land, Sea) > Payload Carrier Platform ✕

Medical, Human Factors & Physiology > Medical Monitoring ✕

Medical, Human Factors & Physiology > Physiological Sensing ✕

Medical, Human Factors & Physiology > Triage Systems ✕

Medical, Human Factors & Physiology > Wound Monitoring ✕

Medical, Human Factors & Physiology > Biomedical Carrier System ✕

Manufacturing & Materials > Additive Manufacturing ✕

Manufacturing & Materials > Composite Materials ✕

Manufacturing & Materials > Fabrication Technologies ✕

Manufacturing & Materials > Polymer Processing ✕

ADDITIONAL TAGS

If you wish to describe your technology with any other tags, please single tags of up to 50 characters, separated by commas.

VTOL, Hybrid, Tilt-rotor, temp-controlled cargo

AGREEMENT TO T&CS * ()

In accordance with section 7.2(a) of the Terms and Conditions.

I confirm I have read the Terms and Conditions of Bidding in full and agree to them.

CONFIRMATION OF COMPANY IDENTITY * ()

This proposal is submitted by a user working for the company under which this proposal is registered; and this is the company that would enter into an agreement with DIANA if selected.

CONFIRMATION THAT COMPANY IS NOT A UNIVERSITY OR NON-PROFIT * ()

I confirm that the entity submitting the proposal is not a university, non-profit or non-for-profit organisation.

PERSONNEL AGREEMENT * ()

I agree that the key personnel involved in the Step 3 - Virtual Panel Session, should that point be reached, will be directly involved in the accelerator programme and the required travel.

ACCELERATOR PROGRAMME AGREEMENT * ()

I understand and accept that the accelerator programme includes approx. 6 in-person mandatory in-person thematic clusters at the assigned accelerator site and event locations involving transatlantic travel. Further I understand and accept that the accelerator programme includes mandatory completion of assignments.

Information Sharing

BIDDER IDENTITY

In accordance with Section 7.2(a) of the Terms and Conditions of Bidding, DIANA may share bidder identity with NATO and Allied stakeholders to foster further opportunities, unless you opt out of such sharing:

BIDDER DOCUMENTATION

In accordance with Section 7.2(b) of the Terms and Conditions of Bidding, DIANA may share bidder documentation with NATO and Allied stakeholders to foster further opportunities, unless you opt out of such sharing:

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